A report on

Geo-Secured QR Code Attendance Platform with Real-Time Proxy-Prevention

Submitted in fulfilment for

Mini-Project I (SY-IT)

Submitted by

Piyush Pawar (246109015)

Atharva Kansurkar(246109025)

Ashish Jaiswal(246109060)

Under the guidance

of

Prof. Varad potdar

Department of Information Technology,

Walchand College of Engineering, Sangli.

Maharashtra, India. 41641

2025-2026

<u>Report on Mini Project</u>

CERTIFICATE

This is to certify that the project work entitled as
"Secure QR-Based Attendance System with Multi-Layer Validation"

Submitted by:

Mr. Piyush Pawar

Mr. Atharva Kansurkar

Mr. Ashish Jaiswal

In fulfillment of the requirement for the degree of

Bachelor of Technology

In

INFORMATION TECHNOLOGY

From

Walchand College of Engineering, Sangli

Affiliated to Shivaji University, Kolhapur

This project work is a record of students' own work carried out by them under my supervision and guidance during the session 2025-26

Prof. Varad Potdar (Project Guide)	(External Examiner)	Dr. R. R. Rathod (HoD)

Dr. A. J. Umbarkar

(Panel Member)

WALCHAND COLLEGE OF ENGINEERING, SANGLI

Sangli - 416415

2025-26

DECLARATION

I hereby declare that the project report entitled **“Secure QR-Based Attendance System with Multi-Layer Validation”** submitted by me to Walchand College of Engineering, Sangli in fulfillment of the requirement for the award of the degree of **Bachelor of Technology in Information Technology** is a record of bonafide project work carried out by me under the guidance of **Prof. Varad Potdar**

I further declare that the work reported in this project report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this Institute or any other institute or university.

I declare that this project report reflects my own understanding of the subject and is written in my own words. I have sufficiently cited and referenced the sources that I have referred to in this work. I have not misrepresented, fabricated, or falsified any idea, data, fact, or source in this submission.

I understand that any violation of the above statements will be subject to disciplinary action by the Institute.

Piyush Pawar (246109015)

Atharva Kansurkar (246109025)

Ashish Jaiswal 246109060)

Date:

Place:

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Prof. Varad Potdar sir** for his valuable guidance, continuous support, and encouragement throughout the development of this project. His suggestions and insights were extremely helpful at every stage of the work.

I would also like to thank **Dr. R. R. Rathod**, Head of the Department of Information Technology, Walchand College of Engineering, Sangli, for providing the necessary facilities and environment to successfully complete this project.

I extend my heartfelt thanks to all the faculty members of the Information Technology Department for their guidance and support whenever required. I also appreciate the help provided by the non-teaching staff.

I am deeply grateful to my parents and family members for their constant motivation, encouragement, and support throughout my academic journey.

Finally, I would like to thank my friends and project teammates for their valuable suggestions, cooperation, and support during the completion of this project.

Piyush Pawar (246109015)

Atharva Kansurkar (246109025)

Ashish Jaiswal 246109060)

ABSTRACT

Attendance management is an essential part of academic institutions, but traditional methods are often time-consuming and prone to errors and misuse such as proxy attendance. To address these issues, this project presents a **Secure QR-Based Attendance System with Multi-Layer Validation**, designed to provide a reliable, efficient, and cost-effective solution.

The proposed system uses dynamically generated QR codes for marking attendance, which change at regular intervals to prevent misuse. To enhance security and ensure authenticity, multiple validation techniques are integrated into the system. These include **QR code encryption**, **geolocation-based verification**, and **device fingerprinting**, ensuring that attendance is marked only by authorized students who are physically present in the classroom.

The system is implemented as an Android application with a cloud-based backend using Firebase services for authentication and data storage. Teachers can create attendance sessions and display QR codes, while students can scan and validate their attendance through the application. The system performs real-time validation before recording attendance, thereby improving accuracy and preventing fraudulent activities.

The results demonstrate that the proposed system is efficient, secure, and user-friendly, making it suitable for real-world deployment in educational institutions. This approach reduces manual effort, saves time, and significantly improves the reliability of attendance tracking.

TABLE OF CONTENTS

1) INTRODUCTION

1.1) Overview of Attendance Systems

Attendance systems are an essential part of educational institutions as they help in tracking the presence of students during lectures. Traditionally, attendance has been marked manually using registers, where teachers call out names and record attendance.

With technological advancements, digital systems such as biometric and RFID-based attendance systems have been introduced. These systems aim to improve efficiency and reduce manual effort. However, despite these improvements, many systems still face issues related to security, cost, and ease of use.

1.2) Need for Smart Attendance System

In current scenarios, traditional attendance methods are either time-consuming or insecure. Manual attendance reduces valuable lecture time, while biometric and RFID systems require additional hardware and infrastructure.

Another major issue is proxy attendance, where one student marks attendance for another. This reduces the reliability of attendance records and creates unfair academic evaluation.

With the widespread use of smartphones, there is a need for a smart attendance system that is:

- Easy to use
- Secure and reliable
- Cost-effective
- Capable of real-time tracking

A mobile-based solution can overcome many of these problems effectively.

1.3) Problem Statement

The major problem addressed in this project is the inefficiency and lack of security in traditional attendance systems.

The key issues include:

- Time-consuming manual attendance process
- Possibility of proxy attendance
- Dependence on expensive hardware in digital systems
- Lack of proper validation of student presence
- No real-time monitoring of attendance

To solve these issues, a system is required that ensures only students who are physically present and authorized can mark their attendance.

1.4) Project Overview

The proposed system is an Android-based application that provides a secure way of marking attendance using QR codes.

In this system, the teacher creates an attendance session and a QR code is generated. This QR code changes dynamically after a fixed interval. Students scan the QR code using their mobile application.

The system verifies multiple conditions such as location, device identity, and QR validity before marking attendance. Once validated, the attendance is stored in a cloud database.

1.5) Scope of the Project

The scope of this project is to develop a secure, efficient, and scalable attendance system for educational institutions.

The system focuses on:

- Replacing manual attendance methods
- Improving accuracy and reliability
- Providing real-time data storage and access

The project can be extended in the future by adding features such as analytics, reporting, and integration with other institutional systems.

2) OBJECTIVES

- To develop a **secure and efficient attendance management system** using mobile technology. The system is designed to overcome the drawbacks of traditional attendance methods and provide a more reliable solution.

- To **eliminate proxy attendance**. The system ensures that only authorized students who are physically present in the classroom can mark their attendance by using multiple validation techniques.

- To **reduce the time required for attendance marking**. Instead of manually calling out names, attendance can be recorded quickly using QR code scanning, which saves valuable lecture time.

- To provide a **cost-effective solution** by avoiding the use of additional hardware such as biometric devices. It makes use of smartphones and cloud services, which are easily accessible.

- To **improving accuracy and transparency** in attendance records. By storing data in real time, teachers can easily monitor and manage attendance.

- Overall, the objective is to create a system that is **secure, fast, easy to use, and suitable for real-world implementation** in educational institutions.

3) EXISTING SYSTEM

3.1) Traditional Attendance Methods

In most educational institutions, attendance is recorded using manual methods such as registers. In this process, the teacher calls out the names of students and marks their presence. Although this method is simple, it is time-consuming and not efficient for large classrooms.

To improve this, some institutions have adopted digital systems like biometric attendance and RFID-based systems. In biometric systems, students mark attendance using fingerprints or facial recognition, while RFID systems use cards that students scan to record their presence.

These systems reduce manual work to some extent, but they still have certain drawbacks related to cost, maintenance, and security.

3.2) Limitations of Existing Systems

Traditional attendance systems have several limitations. Manual attendance consumes valuable lecture time and can become inefficient, especially in large classes. It is also easy for students to mark proxy attendance, which affects the accuracy of records.

Biometric systems require expensive hardware and regular maintenance. They may also face issues like device failure or slow processing when many students use them at the same time.

RFID-based systems can be misused if students share their cards with others. Moreover, these systems do not ensure that the student is actually present inside the classroom.

Due to these limitations, there is a need for a more secure, efficient, and reliable attendance system.

4) PROPOSED SYSTEM

4.1) System Overview

The proposed system is a **mobile-based QR code attendance system** designed to provide a secure and efficient method for marking attendance. In this system, the teacher creates an attendance session using the application, and a QR code is generated for that session.

The QR code is **dynamic**, meaning it changes after a fixed interval of time. This ensures that the QR code cannot be reused or misused through screenshots. Students use their mobile application to scan the QR code displayed by the teacher.

After scanning, the system performs multiple validation checks such as session verification, location verification, device validation, and QR validity. Only if all conditions are satisfied, the attendance is marked successfully and stored in the cloud database.

4.2) Key Features

The proposed system includes several important features that make it more secure and efficient compared to traditional systems.

One of the main features is the use of **dynamic QR codes**, which refresh after a short interval. This prevents misuse of static QR codes.

The system uses **encryption techniques** to secure the QR data, ensuring that it cannot be easily decoded or tampered with by unauthorized users.

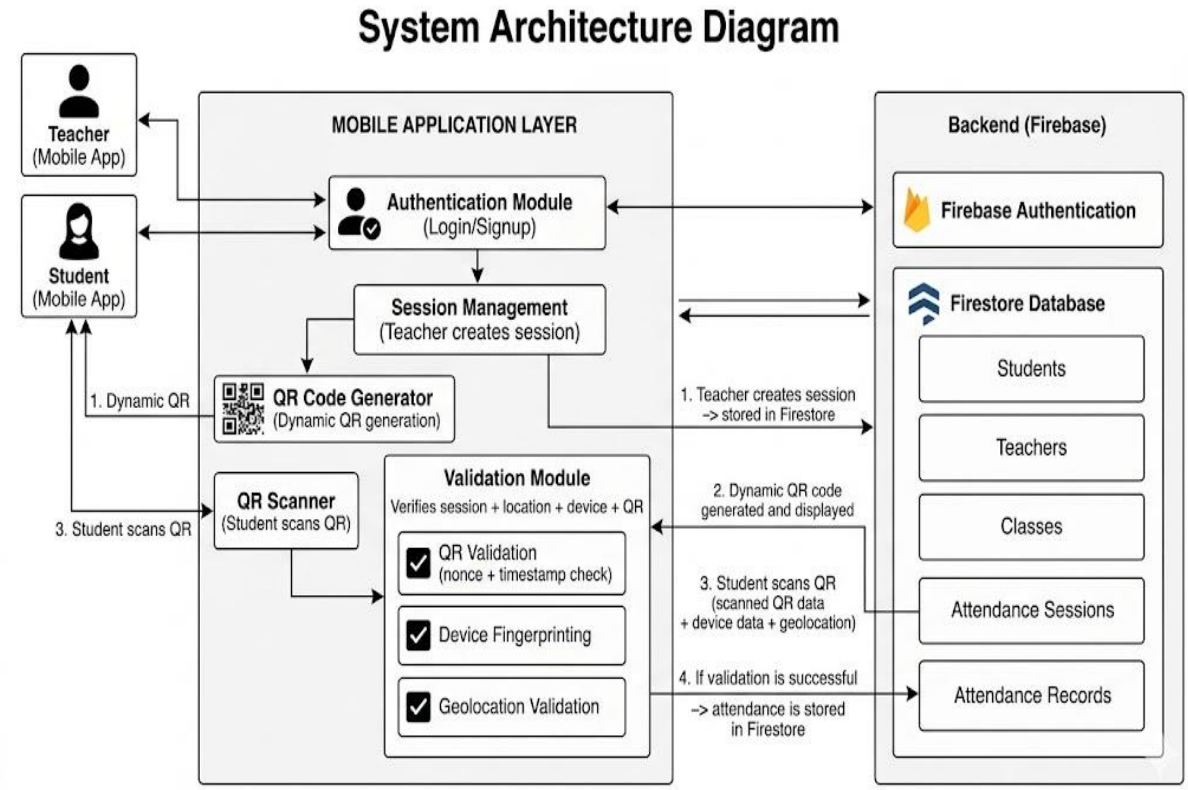
It also includes **geolocation validation**, which checks whether the student is physically present within the classroom area before allowing attendance.

Another important feature is **device-based validation**, which ensures that attendance is marked only from a registered device.

The system also provides **real-time data storage and tracking** using cloud services, allowing teachers to monitor attendance efficiently.

5) SYSTEM ARCHITECTURE

5.1 Architecture Diagram



5.2) System Components

The system is divided into multiple components that work together to perform attendance marking.

- **Mobile Application**

The mobile application is the main interface for both teachers and students. It provides features such as login, session creation, QR display, and QR scanning. It also handles validation processes and user interactions.

- **Firebase Backend**

Firebase is used as the backend system to manage authentication and data storage. It stores information related to users, attendance sessions, and attendance records. It also enables real-time updates and secure data handling.

- **QR Processing Module**

This module is responsible for generating and scanning QR codes. It creates encrypted QR data and updates it dynamically. On the student side, it scans and decodes the QR code for further validation.

- **Validation Module**

This component ensures the security of the system by verifying multiple conditions such as session validity, location, device identity, and QR authenticity before marking attendance.

6) TECHNOLOGIES USED

- The proposed system is developed using modern technologies that ensure efficiency, security, and scalability.

- The **Android platform (Java)** is used for developing the mobile application. It provides a user-friendly interface for both teachers and students and supports functionalities such as QR scanning, session management, and real-time interaction.

- **Firebase Authentication** is used to handle user login and registration. It ensures secure authentication and supports role-based access for teachers and students.

- **Firebase Firestore** is used as the cloud database to store user data, attendance sessions, and attendance records. It provides real-time data synchronization and efficient data management.

- For QR code functionality, the system uses two technologies. The **ZXing (Zebra Crossing) library** is used for generating QR codes on the teacher side. It allows efficient and customizable QR code creation. For scanning QR codes on the student side, **CameraX along with Google ML Kit** is used, which provides accurate and fast real-time QR detection.

- To enhance security, **AES encryption** is used to encrypt the QR code data. This ensures that the data cannot be easily decoded or tampered with.

- The system also uses **Location Services (GPS)** to verify whether the student is physically present within the classroom area before marking attendance.

- These technologies together help in building a secure, efficient, and reliable attendance system.

7) METHODOLOGY

The methodology describes the step-by-step working of the proposed attendance system. It explains how different components interact to perform secure attendance marking.

7.1) User Authentication

The system begins with user authentication. Both teachers and students are required to log in using their credentials.

Firebase Authentication is used to verify user identity. Based on the login, the system identifies the role of the user (teacher or student) and provides access to the respective features.

This ensures that only authorized users can access the system.

7.2) Session Creation

After logging in, the teacher creates an attendance session using the application.

The session includes important details such as:

- Class
- Subject
- Location of the classroom

Once the session is created, it is stored in the cloud database and becomes active for students to mark attendance.

7.3) QR Code Generation

After creating the session, a QR code is generated dynamically.

The QR code contains encrypted data such as:

- Session ID
- Timestamp
- Nonce (unique value)

The QR code refreshes at regular intervals (e.g., every 10 seconds) to prevent misuse. The previous QR code remains valid for a short duration to handle delays during scanning.

7.4) QR Code Scanning

Students use their mobile application to scan the QR code displayed by the teacher.

The QR scanner uses the device camera to detect and decode the QR code. Once scanned, the encrypted data is decrypted within the application for further processing.

7.5) Validation Process

After scanning the QR code, the system performs multiple validation checks to ensure security.

These include:

- **Session Validation** – checks if the session is active
- **QR Validation** – verifies the nonce and timestamp
- **Device Validation** – checks if the device is registered
- **Location Validation** – verifies if the student is within the classroom area

Only if all these conditions are satisfied, the system allows attendance marking.

7.6) Attendance Marking

Once the validation is successful, the attendance is marked for the student.

The system records:

- Student ID
- Session ID
- Timestamp
- Location

This data is stored in the cloud database in real time. If any validation fails, the attendance is rejected and an appropriate message is shown.

8) SECURITY MECHANISMS

The proposed system implements multiple layers of security to ensure that attendance is marked only by legitimate students. Each mechanism addresses a specific type of misuse and together they provide a robust and reliable system.

8.1) Dynamic QR Code (Nonce)

The system uses a **dynamic QR code mechanism** where the QR code changes automatically after a fixed interval (approximately every 10 seconds). Each QR code contains a unique value known as a **nonce**, along with a timestamp and session information.

The nonce is generated randomly and is valid only for a short duration. When a new QR code is generated, the previous QR code is also stored temporarily with a limited validity period (grace window). This is done to handle practical delays such as slow scanning or location fetching.

The use of dynamic QR codes prevents common attacks such as:

- Using screenshots of QR codes
- Sharing QR codes among students
- Reusing old QR codes

During validation, the system checks whether the scanned nonce matches the current or recently expired nonce within the allowed time window. If the nonce is invalid or expired, the attendance is rejected.

Thus, the dynamic QR code mechanism ensures that only students who are present during the active session can mark attendance.

8.2) AES Encryption

To secure the QR code data, the system uses **AES (Advanced Encryption Standard)** encryption. The QR code does not store plain text information; instead, all important data such as session ID, timestamp, and nonce are encrypted before generating the QR code.

AES is a symmetric encryption algorithm that provides strong security and is widely used in modern applications. A unique session key is generated for each attendance session and is used to encrypt and decrypt QR data.

This ensures that:

- QR code data cannot be read directly by scanning apps
- Unauthorized users cannot modify or forge QR data
- Sensitive information remains protected

Even if someone scans the QR code using an external application, the data appears meaningless without the correct decryption key.

8.3) Device Fingerprinting

The system uses **device fingerprinting** to uniquely identify each student's device. It collects various device-specific parameters such as Android ID, device hardware details, and application signature to generate a unique fingerprint.

This fingerprint is stored in the database and is linked to the student account. The system allows attendance marking only from registered devices.

Key benefits of this approach:

- Prevents multiple device usage for a single account
- Reduces chances of account sharing
- Detects suspicious login behavior

Additionally, the system can restrict the number of devices allowed per user (for example, maximum two devices), making it more secure while still flexible.

8.4) Geolocation Validation

The system uses **GPS-based geolocation validation** to ensure that the student is physically present in the classroom.

When the teacher creates a session, the location (latitude and longitude) of the classroom is stored. A **geofence** is created by defining a radius around this location.

During attendance marking, the student’s current location is fetched using GPS. The system then calculates the distance between the student’s location and the classroom location.

If the student is within the allowed radius, the validation passes; otherwise, the attendance is rejected.

Additional checks include:

- Location accuracy validation
- Detection of mock or fake locations
- Freshness of location data

This ensures that students cannot mark attendance from outside the classroom.

9) DATABASE DESIGN

The system uses **Firebase Firestore (NoSQL database)**

- Stores data in **collections and documents**
- Supports **real-time updates**
- Easy to scale and manage

Students Collection

Stores student details:

- Name → used for identification
- Roll number → unique student ID
- Class → links student to class
- Email → used for login/authentication
- Device ID → used for device verification
- FCM token → used for notifications

Teachers Collection

Stores teacher information:

- Name → teacher identification
- Email → login/authentication
- Subject → subject handled
- Classroom → default class location
- FCM token → for sending notifications

Classes Collection

Stores class-related data:

- Class name → identifies class
- Subject → subject assigned
- Teacher ID → links to teacher
- Enrolled students → list of student IDs

Attendance Sessions Collection

Main collection for sessions:

- Session ID → unique session identifier
- Class ID → links session to class
- Teacher ID → identifies session owner
- Current QR (nonce) → active QR value
- Previous QR → used for grace period
- Session key → used for encryption
- Location (lat, long) → classroom location
- Geofence radius → allowed range
- Start time → session start
- Status → active/inactive

Attendance Records (Subcollection)

Stored inside each session:

- Student ID → who marked attendance
- Session ID → which session
- Timestamp → time of marking
- Device ID → device verification
- Location → student location
- Status → present/rejected
- Rejection reason → if validation fails

Data Model — Firestore Collections

```
Firestore Root
│
├── students (collection)
│   └── {studentId}
│   ├── name: "Rahul Sharma"
│   ├── rollNo: "CS21"
│   ├── class: "TY BSc CS"
│   ├── email: "rahul@gmail.com"
│   ├── phone: "9876543210"
│   ├── deviceId: "abcd1234" / primary device fingerprint
│   ├── deviceId2: "" / optional 2nd device
│   └── fcmToken: ". "
│
├── teachers (collection)
│   └── {teacherId}
│   ├── name: "Mr. Patil"
│   ├── email: "patil@college.edu"
│   ├── subject: "Data Structures"
│   ├── classroom: "Room 301"
│   └── fcmToken: ". "
│
├── classes (collection)
│   └── {classId}
│   ├── className: "TY BSc CS"
│   ├── subject: "DSA"
│   ├── teacherId: "teacherId_001"
│   └── enrolledStudents: [studentId_001, . ]
│
└── attendanceSessions (collection)
└── {sessionId}
├── classId: "class_001"
├── teacherId: "teacherId_001"
├── qrCode: "RANDOM_TOKEN_12345" / current nonce
├── latitude: 19.9975
├── longitude: 73.7898
├── geofenceRadius: 50
├── startTime: timestamp
├── endTime: timestamp
├── active: true
└── records (subcollection)
└── {studentId}
├── status: "present" | "rejected"
├── time: timestamp
├── deviceId: "abcd1234"
├── deviceLocationLat: 19.9976
├── deviceLocationLong: 73.7897
└── rejectionReason: "" | "location_mismatch" | .
```

10) IMPLEMENTATION DETAILS

The system is implemented as an **Android application** with Firebase as the backend. The implementation is divided into different modules, each responsible for a specific functionality.

10.1 Authentication Module

Handles user login and registration:

- Firebase Authentication used
- Supports email/password login
- Identifies user role (Student/Teacher)
- Provides secure access to system

10.2 Student Module

Used by students to mark attendance:

- Scan QR code using camera
- Decrypt QR data
- Perform validation checks
- View attendance status/history

10.3 Teacher Module

Used by teachers to manage sessions:

- Create attendance session
- Generate dynamic QR code
- Display QR code to students
- Monitor attendance records

10.4 QR Generation Module

Handles QR creation:

- Uses **ZXing library**
- Generates QR from encrypted data
- Includes session ID, timestamp, nonce
- QR refreshes every few seconds

10.5. QR Scanning Module

Handles QR scanning:

- Uses **CameraX + ML Kit**
- Detects QR code from camera
- Decodes encrypted data
- Passes data for validation

10.6. Validation Module

Ensures security before marking attendance:

- Session validation
- QR (nonce + time) validation
- Device fingerprint verification
- Location (geofence) check

10.7. Database Module

Handles data storage:

- Firebase Firestore used
- Stores users, sessions, records
- Real-time updates enabled

10.8 Implementation Flow

- User logs in → role identified
- Teacher creates session → stored in database
- QR code generated and displayed
- Student scans QR → data decoded
- System performs validations
- If valid → attendance stored
- If invalid → rejected

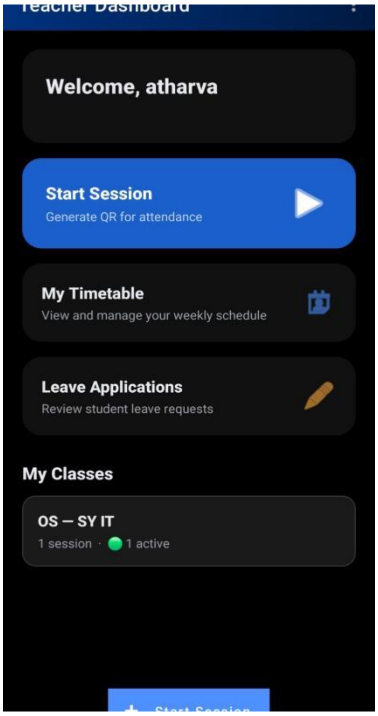
11) RESULTS AND SCREENSHOTS

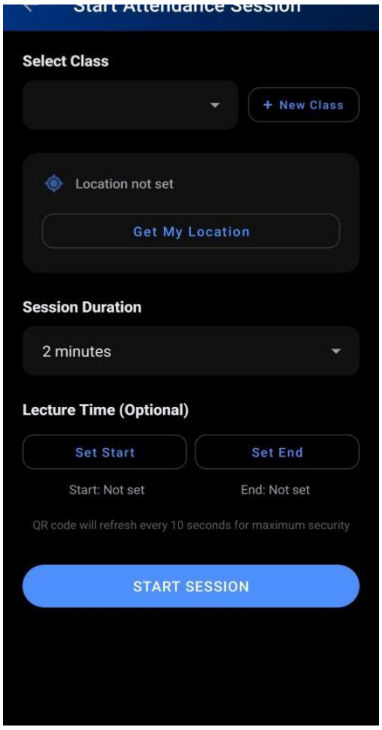
The proposed system was successfully implemented and tested on Android devices. All the core functionalities such as login, session creation, QR generation, QR scanning, validation, and attendance marking were working as expected.

Observed Results

- User authentication works correctly for both students and teachers .
- Teacher is able to create attendance sessions successfully .
- Dynamic QR code is generated and refreshes automatically .
- Students are able to scan QR codes with some delay due to multiple times location fetch for accuracy.
- QR data is decrypted and processed correctly
- Validation checks (session, device, location, QR) are working properly
- Attendance is marked only when all validations pass
- Invalid attempts are rejected with proper reasons

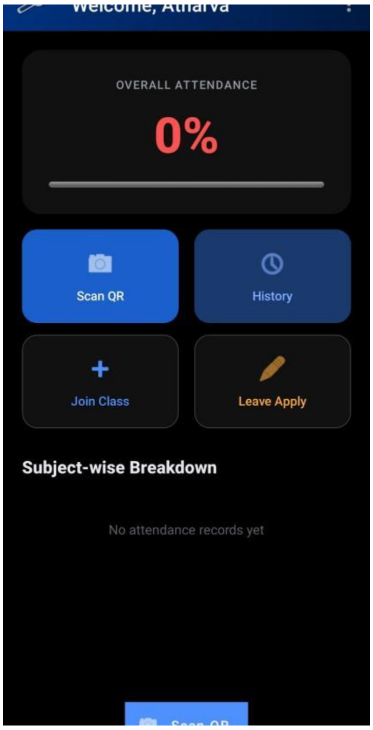
Teacher Dashboard 1



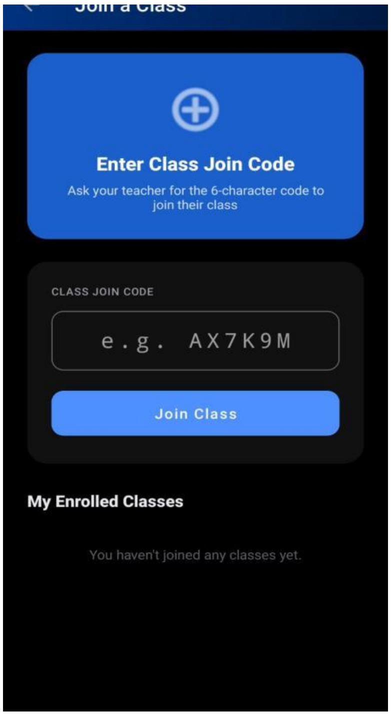


Teacher Dashboard

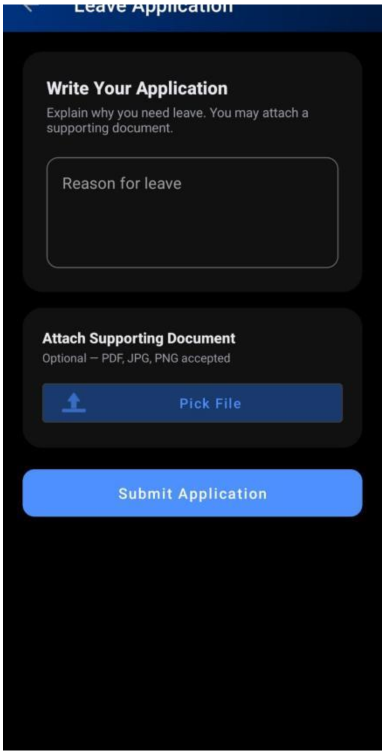
QR generation page



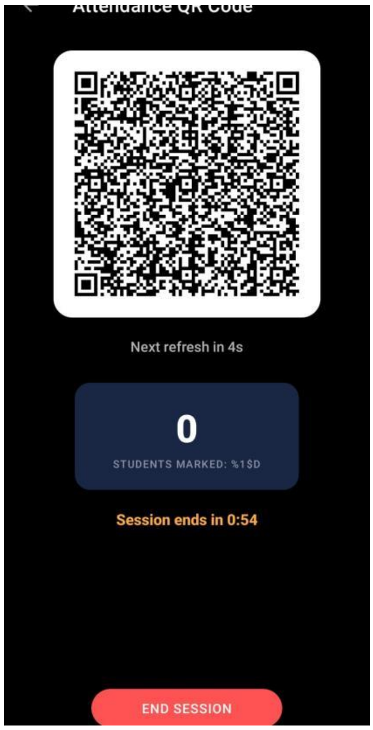
Student Dashboard



Join Class page



Student Leave Application



QR Display

12) ADVANTAGES

- **Prevents Proxy Attendance**

Uses multiple validation layers such as QR, device, and location → ensures only genuine students can mark attendance

- **Saves Time**

Eliminates manual roll calls → attendance marked quickly through QR scanning → increases teaching time

- **No Extra Hardware Required**

Works completely on smartphones → no need for biometric devices or RFID systems → reduces cost

- **Real-Time Data Storage**

Attendance stored instantly in cloud → teachers can monitor live data → no delay in record updates

- **High Security**

Uses encryption, dynamic QR, and validation checks → prevents misuse and data tampering

- **User-Friendly System**

Simple interface for both students and teachers → easy to operate without technical knowledge

- **Scalable Solution**

Can be used in large classrooms and institutions → supports multiple users and sessions

- **Accurate Record Management**

Reduces human errors → ensures proper tracking of attendance

13) LIMITATIONS

- **Internet Dependency**

Requires active internet connection, so the system may not work properly in offline conditions.

- **GPS Accuracy Issues**

Indoor environments may reduce GPS accuracy, leading to slight location errors during validation.

- **Device Dependency**

Requires smartphones for operation, so users without compatible devices may face difficulty.

- **Initial Setup Required**

Needs user registration, device binding, and initial configuration before usage.

- **Battery Consumption**

Continuous use of camera and GPS may lead to higher battery usage.

- **Performance Variations**

System performance depends on device quality, and older devices may respond slower.

- **Network Delay Issues**

Slow internet connection can affect QR scanning speed and validation process.

14) FUTURE SCOPE

- The system can be enhanced by integrating **face recognition technology** to verify student identity along with QR scanning, ensuring higher security and eliminating proxy attendance.

- The system can be improved with **high-precision indoor positioning techniques** such as Wi-Fi-based location tracking or BLE triangulation to overcome GPS limitations.

- An **offline attendance feature** can be introduced, allowing temporary data storage and synchronization when an internet connection becomes available.

- The system can be expanded to support **cross-platform applications**, including iOS and web-based interfaces, increasing accessibility.

- Integration of a **notification system** can help in sending alerts for low attendance, reminders for classes, and attendance confirmations.

- Advanced **AI and machine learning techniques** can be used to predict attendance patterns and identify irregular student behavior.

- The system can be integrated with existing **college ERP systems** to manage student records, results, and academic activities in a unified manner.

15) CONCLUSION

The proposed QR-based attendance system provides an effective and modern solution to overcome the limitations of traditional attendance methods used in educational institutions. Conventional systems such as manual attendance, biometric devices, and RFID-based systems either consume significant time or require additional hardware and still fail to completely prevent proxy attendance. This project successfully addresses these issues by introducing a secure, mobile-based attendance system that is easy to use and cost-effective.

The system uses dynamic QR codes that change at regular intervals, which makes it difficult to misuse captured or shared QR codes. In addition to this, multiple validation techniques such as QR verification, device identification, and geolocation checking are implemented to ensure that only authorized students who are physically present in the classroom can mark their attendance. The use of AES encryption further enhances the security of the system by protecting the QR data from unauthorized access or modification.

Another important aspect of the system is the use of Firebase services, which provide secure authentication and real-time cloud storage. This allows attendance data to be stored instantly and accessed easily by teachers, improving transparency and management. The system reduces manual effort, minimizes errors, and saves valuable lecture time, making the overall attendance process more efficient.

The application has been successfully implemented and tested, and it performs all the required operations such as session creation, QR generation, QR scanning, validation, and attendance marking accurately. The system also handles invalid cases properly by rejecting unauthorized attempts, which ensures reliability.

Overall, the project demonstrates that a secure and efficient attendance system can be developed using existing technologies without the need for additional hardware. It is suitable for real-world implementation in colleges and institutions. With further improvements such as offline support, advanced analytics, and system integration, the application can be expanded into a complete attendance management solution.